Isha Mishra

Senior Software Engineer, San Francisco, CA

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Now: Data Fellow, Chicago Education Advocacy Cooperative. Open to Senior Software Engineer roles, Bay Area, CA. Work authorized for 3 years (STEM OPT).

SUMMARY

Senior software engineer with 5 years of backend and platform experience across Oracle, WNS and startups, now building retrieval systems and AI agents with deterministic guardrails.

EXPERIENCE

AI Engineer Intern, Quotr, Berkeley, CA Nov 2025 – Mar 2026

Construction estimation and procurement platform for GCs, backed by Berkeley SkyDeck; seed startup (raised $4M)

- Built hybrid retrieval for construction plans, using OCR on PDF drawings, metadata-based chunking by sheet, trade and system hierarchy, and FAISS + BM25 hybrid search, improving context recall from 0.68 to 0.88.
- Built precision-ranking for estimator review, using confidence scoring on retrieved chunks, metadata-based deduplication, and top-k reranking to improve context precision@10 from 0.61 to 0.84.

Senior Software Engineer, LetsTransport, Bangalore, India Jul 2022 – Nov 2023

End-to-end logistics services for enterprises, e-commerce and retail; Series E startup (raised $84M)

- Built a driver verification triage workflow, using an OPA/Rego decision contract and RabbitMQ notifications to drivers, routing missing or low-confidence cases for human review, to achieve p90 onboarding TAT under 24 hours.
- Integrated MongoDB with Elasticsearch via Monstache for near real-time sync, keeping MongoDB as the single write path and ES as a secondary search index for a selected query set, bringing query latency under 300 ms.
- Built a replica-read path for the driver-allocation service to cut latency during dispatch peaks, moving geospatial reads off the primary DB to a same-AZ replica, capping and pooling connections per pod, and guarding reads with a replica-lag threshold, to reduce p99 latency under 400 ms.
- Built the allocation matching engine, filtering drivers on vehicle type, capacity and availability, selecting a scored pool, and broadcasting concurrently with atomic first-acceptance, to allocate ~80k orders a month.

Senior Software Engineer, WNS Global Services, Bangalore, India Aug 2021 – Jun 2022

Customer Insights Platform, BFSI vertical

- Led end-to-end development of a self-serve customer insights platform for BFSI business users with Spring Boot and Angular, replacing manual report preparation and cutting reporting effort by 40%.
- Architected and optimized RESTful APIs, reducing average response time by 30% across multiple client-facing applications.
- Built automated build-and-deploy pipelines with Jenkins and Docker, containerising services into versioned images and gating releases on automated tests, to cut release lead time from 7 days to 2.

Senior Software Engineer, Oracle, Bangalore, India Jan 2019 – Aug 2021

Oracle Primavera Cloud risk engine for F500 Construction & Engineering (fast-track promotion in 15 months)

- Built weather-data idempotent import in the risk register with Kafka, keying versioned upserts on riskID and reconciling row counts on every run, favouring correctness over freshness, to keep p95 register sync lag under 10 minutes.
- Built the Monte Carlo rollout for risk simulation, distributing per-worker seeding for reproducible sampling and reducing results without shared mutable state, to batch-schedule ~10k activities at ~10k runs per analysis.
- Built qualitative risk matrices, computing pre- and post-response scores against probability and impact thresholds and normalising scores for program-level rollup, with zero gaps or double counting per program.
- Parallelized the Monte Carlo risk engine by chunking iterations across threads, seeding each iteration and merging per-thread results, achieving ~1.4x analysis throughput with results identical to sequential runs.
- Designed and implemented the weather rules engine in the Risk Module: definitions and validation, mapping weather events to non-working days, and configs by region and season.

SELECTED PROJECTS

- **tira** (with Devansh Pathak): A multiplayer task board for AI agents and the people they work with: nothing is marked done without proof (a real commit or pull request, checked on GitHub), and every move is written once to a log nobody can rewrite. github.com/devpath56/tira
- **DebugAssistAgent**: Writes a failing test, finds the cause, fixes it, and guards the whole bug class. You approve before anything posts. github.com/ishamishra0408/DebugAssist
- **DogMaAI**: A home-camera companion: how a dog felt alone, how long it cried, and whether calm music helped. github.com/ishamishra0408/DogMaAI
- **Agent KYCReady**: Driver KYC rebuilt AI-first: a model reads the documents, a Rego policy decides, people take the hard cases. github.com/ishamishra0408/Agent-KYC
- **No-Leak-MCP** (with Devansh Pathak): One poisoned Slack message can make an assistant leak your keys through a link preview. This blocks it before it leaves. github.com/ishamishra0408/NoLeakMCP

SKILLS

Languages: Java, JavaScript, TypeScript, Python, Kotlin, C++

Technologies: Zookeeper, Kafka, Redis, Jenkins, Docker, Kubernetes, GCP, MongoDB, Postgres, Oracle DB

Frameworks: Spring Boot, Node.js, NestJS, JUnit, Angular, REST, SOAP, FastAPI, GraphQL, Integration

AI systems: RAG, FAISS, BM25, OCR, MCP, OPA / Rego, Evals

EDUCATION

M.S. Computer Science, Westcliff University, San Francisco, GPA 4.0 / 4 2024 – 2026

B.Tech, Electrical, Electronics & Communication Engineering, Vellore Institute of Technology (VIT), India, GPA 3.83 / 4 2015 – 2019